

Agenda

1. Announcements

- 11.2 and 11.3 HW due on Wednesday
- Discussion Quiz 7 on Thursday over
- Midterm 2 is on Tuesday April 11 at 7:30pm
 - Check for conflicts and sign up for the makeup if needed
 - Sign up to take exam at the testing center if you have testing accommodations

2. Section 11.5 - Alternating Series

Activity: What test would you use to determine if each Series converges or diverges?

$$1.) \sum_{n=5}^{\infty} \frac{6+8n+9n^2}{3+2n+n^2}$$

$$4.) \sum_{n=2}^{\infty} \frac{1}{n \ln n}$$

$$2.) \sum_{n=1}^{\infty} \frac{3}{n^2+7n+12}$$

$$5.) \sum_{n=1}^{\infty} \left(\frac{1}{n^2} + 1 \right)^2$$

$$3.) \sum_{n=2}^{\infty} \frac{5^{n+1}}{7^{n-2}}$$

$$6.) \sum_{n=2}^{\infty} \frac{7}{n(n+1)}$$

Alternating Series

Examples: The following are all alternating series:

1.) $\sum_{n=1}^{\infty} (-1)^{n-1} \Rightarrow 1 - 1 + 1 - 1 \dots$ (not 0) ... 2.) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \Rightarrow 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

3.) $\sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n \Rightarrow 1 - \frac{2}{3} + \frac{4}{9} - \frac{8}{27} + \dots$ 4.) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -1 + \frac{1}{4} - \frac{1}{9} + \frac{1}{16} - \dots$

'Def': whenever the sign flips

Definition:

An alternating series is a series which can be written in the form: $\sum_{n=1}^{\infty} (-1)^{n-1} a_n = a_1 - a_2 + a_3 - \dots$

OR $\sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + a_4$ where $a_n \geq 0$ for $n \geq 1$

Definition: A series is said to be absolutely convergent if the $\sum |a_n|$ converges.

Activity: Determine whether each series is absolutely convergent.

1.) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n^2} \right| \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges because $p=2 > 1$, so $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ is absolutely convergent

2.) $\sum_{n=1}^{\infty} \left| \left(-\frac{2}{3}\right)^n \right| = \sum_{n=1}^{\infty} \left| (-1)^n \left(\frac{2}{3}\right)^n \right| = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ absolutely converges

3.) $\sum_{n=1}^{\infty} \left| \frac{1}{n^{1.01}} \right|$ $p > 1$, absolutely converges

4.) $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n} \leftarrow$ diverges $\left| \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \right|$ not absolutely convergent.

Theorem: If $\sum a_n$ absolutely converges
then $\sum a_n$ converges.

This tells us $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges

↙ "alternating harmonic series"

Does $\sum \frac{(-1)^{n-1}}{n}$ diverge or converge?
need new test.

Alternating Series Test:

Suppose $a_n \geq a_{n+1}$ for all $n \geq 1$, and $a_n \geq 0$ for all $n \geq 1$, with $\lim_{n \rightarrow \infty} a_n = 0$.
Then the series $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$ converges

Example: Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converges

decreasing? $\frac{1}{n} \geq \frac{1}{n+1}$ $n+1 > n$ ✓

positive? $\frac{1}{n} \geq 0$ yes ✓ therefore $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$
converges based on
the alternating
series test.

$\lim_{n \rightarrow \infty} a_n = 0$? $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ ✓

Activity: Does $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+4)}$ converge?

Definition:

Estimating the Sum Using Alternating Series Test

Remainder Estimation

Example: Estimate the sum of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ using S_{10}